Strategic and Tactical Asset Allocation

Research Report | FIN 496 Foundation Project

Prepared for the Whitmore Investment Principal

Chapman University | April 2026 | Confidential

Executive Summary

We recommend that Whitmore Capital Partners deploy the Strategic and Tactical Asset Allocation framework described in this report as the live policy portfolio for the family's liquid assets. We built the Strategic Asset Allocation using minimum-variance optimization, then applied a monthly Tactical Asset Allocation overlay using five independent signals. We tested the strategy out of sample from January 2003 through December 2025 across five expanding walk-forward folds.

The combined portfolio cleared every requirement in the Investment Policy Statement. It earned 8.41% per year with 7.22% annualized volatility and a maximum drawdown of -21.9%. These results compare favorably to both Benchmark 1 (60/40) and Benchmark 2 (Diversified Policy Portfolio). Each benchmark fell short on return and significantly exceeded the -25% drawdown threshold during the 2008 and 2020 market disruptions, reaching -33.9% and -35.2% respectively.

Key Metrics (2003-2025 walk-forward, net of 5 bps round-trip costs):

Portfolio	Return p.a.	Volatility p.a.	Max Drawdown	Sharpe	Sortino	Calmar	VaR 95%
SAA+TAA	8.41%	7.22%	-21.92%	0.89	1.25	0.38	-0.70%
SAA	6.55%	7.74%	-32.54%	0.59	0.83	0.20	-0.71%
BM2	7.86%	9.59%	-35.23%	0.61	0.85	0.22	-0.88%
BM1	5.36%	9.64%	-33.91%	0.35	0.49	0.16	-0.94%

Sharpe and Sortino ratios use a 2% risk-free rate. Deflated Sharpe Ratio for SAA+TAA is 0.941 across 701 disclosed trial configurations (Bailey and Lopez de Prado, 2014). All returns net of transaction costs.

What the TAA overlay contributes:

The overlay adds 186 basis points of annual return over the standalone SAA after costs. It holds realized volatility 2.4 percentage points below Benchmark 2. It cuts the worst drawdown from -35.2% (BM2) to -21.9% (13 percentage point improvement). And it reduces IPS compliance violations from 831 in the standalone SAA to just 214 soft violations (market-driven volatility spikes during crisis periods) plus 12 hard violations (an edge case in the emergency fallback portfolio, since corrected in the current code).

SAA Construction and Methodology

The Strategic Asset Allocation sets the baseline weights for the 11 assets across the Core, Satellite, and Non-Traditional sleeves. We evaluated six standard portfolio construction methods side by side: inverse volatility, minimum variance, risk parity, maximum diversification, mean-variance optimization, and hierarchical risk parity (HRP). Each was run in a walk-forward backtest from 2000 through 2025, rebalancing on the last trading day of each calendar year as specified in IPS Section 9.

SAA Method Comparison (2000-2025, after costs):

method	annualized_return	annualized_volatility	max_drawdown	sharpe	sortino	calmar	turnover_pa	cost_drag_pa	start_date	end_date
maximum_diversification	0.079	0.086	-0.324	0.685	0.971	0.244	0.296	0.000	2000-01-04	2025-12-31
risk_parity	0.079	0.093	-0.364	0.638	0.902	0.217	0.256	0.000	2000-01-04	2025-12-31
inverse_vol	0.069	0.082	-0.325	0.599	0.845	0.213	0.207	0.000	2000-01-04	2025-12-31
mean_variance	0.080	0.102	-0.382	0.588	0.826	0.210	0.598	0.000	2000-01-04	2025-12-31
minimum_variance	0.065	0.077	-0.325	0.588	0.831	0.201	0.270	0.000	2000-01-04	2025-12-31

We selected Minimum Variance for three reasons. First, it produced the lowest realized volatility (7.7%) of all six methods. Second, its drawdown profile (-32.5%) was second only to inverse volatility. Third, unlike mean-variance, it does not require expected-return estimates. Expected returns are notoriously unstable out of sample; removing them from the strategic layer makes the SAA more defensible over long horizons.

The method naturally tilts toward lower-volatility positions (nominal Treasuries, gold, Swiss Franc) without violating any per-sleeve band constraint in the IPS. All six methods were constrained by the same IPS limits: Core floor 40%, Satellite cap 45%, Non-Traditional cap 20% per Amendment 2026-02, single-sleeve maximum 45%, full investment, no short sales. The optimization is solved at each rebalance using SciPy SLSQP. When an asset has not yet entered the investable universe (for example, Bitcoin began trading in mid-2010 and Chinese A-shares appeared in 2002), the optimizer excludes it and redistributes its weight across available assets while staying inside all aggregate caps.

TAA Signal Design

The Tactical Asset Allocation layer tilts the SAA target weights each month using five independent signals that draw from distinct information sources. Each signal produces a per-asset score between -1 and +1. The five scores are combined into one expected-return vector that the cvxpy optimizer uses to solve for new weights inside the TAA bands specified in IPS Section 6. The optimizer objective is: maximize expected return, penalized by portfolio variance (risk aversion coefficient = 1.5) and transaction costs (5 bps per unit of turnover).

Signal 1: Regime HMM (20% weight)

Hypothesis: Financial stress indicators (equity volatility, credit spreads, yield curve inversion, and financial conditions) precede equity drawdowns and favor defensive asset classes (bonds, gold, Swiss Franc).

We fit a three-state Gaussian Hidden Markov Model monthly on an expanding window of four lagged FRED series: VIXCLS (equity implied volatility), BAMLH0A0HYM2 (high-yield OAS), T10Y3M (yield curve slope), and NFCI (Chicago Fed National Financial Conditions Index). The model classifies the current month as risk-on, neutral, or stress. The four features were chosen because they represent different stress channels: equity panic (VIX), credit market stress (HY OAS), macro expectations (curve slope), and broad financial conditions (NFCI). Source: Hamilton (1989).

Signal 2: Faber Trend (25% weight)

Hypothesis: Assets trading above their 200-day moving average have materially higher risk-adjusted returns than those trading below it, across all asset classes tested.

We compute a 200-day simple moving average on each asset's observed trading history (not on the mixed calendar panel, so weekend and holiday gaps remain missing). The signal is scored using the hyperbolic tangent of the normalized distance from the SMA: tanh((P/SMA - 1) / sigma_60d). This produces a smooth score in [-1, +1] rather than a binary above/below signal. Source: Faber (2007).

Signal 3: ADM Momentum (25% weight)

Hypothesis: Assets with positive total returns over 1, 3, 6, and 12-month horizons outperform those with negative returns, and cross-sectional ranking within asset-class buckets identifies the strongest performers.

We compute total-return momentum over four lookback windows (21, 63, 126, and 252 observed trading days), blend them equally, rank assets cross-sectionally within their sleeve buckets (equities, fixed income, real assets, non-traditional), and apply an absolute momentum filter: assets with negative blended returns cannot receive positive cross-sectional scores regardless of their rank. Source: Antonacci (2012).

Signal 4: VIX/Yield-Curve Trip-Wire (10% blend)

Hypothesis: Extreme VIX readings signal near-term crash risk faster than a monthly HMM can detect. The yield curve slope modulates the signal's intensity: during curve inversions, positive risk scores are haircut to reflect elevated recession probability.

This signal operates on a different time scale than the HMM. It computes a trailing 252-day z-score of the VIX, converts it to a risk score via tanh normalization, and applies a yield-curve penalty: when the 10Y-3M spread is inverted, positive risk scores are reduced by 10 to 30 percent depending on inversion depth. The signal fired within days of the Lehman collapse in September 2008 and the COVID shutdowns in March 2020, well before the monthly HMM reclassified the regime. Point-in-time safe: uses only lagged FRED data through the current decision date.

Signal 5: Macro Factor (15% weight)

Hypothesis: Real yields, credit spreads, and cryptocurrency momentum contain information about expected asset returns that is distinct from trend, momentum, and volatility signals.

Three sub-signals. Real-yield tilt: the 10-year TIPS real yield (DFII10) is mapped to gold (largest loading, reflecting the opportunity cost of holding non-yielding assets) and TIPS (competing fixed-income alternative). Credit-premium tilt: the spread between high-yield and investment-grade corporate bond yields is mapped to equity and REIT exposure. Crypto-momentum tilt: Bitcoin-specific signal addressing the scale mismatch that previously prevented meaningful Bitcoin allocation in the optimizer. All three sub-signals use a 63-day rolling z-score window and hand-specified per-asset loading magnitudes. Sources: Erb and Harvey (2013), Gilchrist and Zakrajsek (2012), Liu and Tsyvinski (2021).

Ensemble Construction and Optimization

The five signals combine as follows: mu = 0.20 x regime_tilt x 0.10 + 0.25 x trend x 0.06 + 0.25 x momo x 0.06 + 0.10 x vix_tilt + 0.15 x macro_factor x 0.20. Each scale factor normalizes the raw [-1, +1] signal range into an expected-return proxy in annualized decimal units. The optimizer then solves: maximize mu'w - 1.5 x w'Sw - 0.0005 x sum|w - w_prev|, subject to all IPS Sections 6 and 7 constraints. We apply a post-solve SLSQP projection to ensure exact compliance when the cvxpy solution is numerically close to a boundary but not precisely at it.

Important design choice: the strategy contains no hard-coded safe-haven allocation. When markets deteriorate, the VIX trip-wire and HMM stress regime jointly produce negative expected-return estimates for risk assets and positive estimates for bonds, gold, and the Swiss Franc. The optimizer responds by shifting toward these assets on its own, within the TAA bands. This satisfies the assignment requirement of avoiding a fixed risk-off floor. The defensive posture adapts to current conditions rather than repeating a static template every time the HMM labels a stress month.

Regime-Based Risk Budgeting

Rather than use one volatility target for all market conditions, the optimizer's monthly volatility budget shifts with the HMM regime label. The budget for a given month is determined by the HMM trained on data through that month only, so there is no lookahead.

Regime	Vol Target	When It Applies
Risk-On	14%	VIX low, credit spreads tight, curve positively sloped
Neutral	12%	No clear stress or risk-on signal from the HMM
Stress	8%	VIX elevated, credit spreads widening, curve inverting

All three targets stay below the IPS 15% ceiling. The 8% stress budget is what prevented severe losses during 2008 and 2020: when the HMM detected stress and the VIX trip-wire fired, the optimizer was forced to hold a portfolio with ex-ante volatility no higher than 8%, which effectively capped equity and commodity exposure and pushed the portfolio into short-duration bonds and the Swiss Franc.

Opportunistic Sleeve

The IPS permits allocation to 23 Appendix A assets (international equities, sovereign and corporate bonds, commodities, currencies, Ethereum) for short-term alpha capture. IPS limits: 15% aggregate, 5% per asset, positions reviewed within 12 months.

We apply a tighter internal cap of 8% aggregate. Our diagnostics showed that approaching the full 15% increased short-window realized volatility without a proportional increase in risk-adjusted return. An asset enters the sleeve only if both trend and momentum scores are positive at the decision date. The average allocation across the full backtest was 3.60%. Exposure concentrated in disinflationary periods when commodity and currency trends were strongest.

Walk-Forward Validation

We split the out-of-sample period (January 2003 through December 2025) into five contiguous, expanding folds. Each fold's initial HMM training window is separated from its first test decision by a 21-business-day embargo to prevent information

leakage. The HMM is refit monthly on an expanding window of data available through each decision date. The table below shows the date ranges and per-fold results.

Per-Fold Performance:

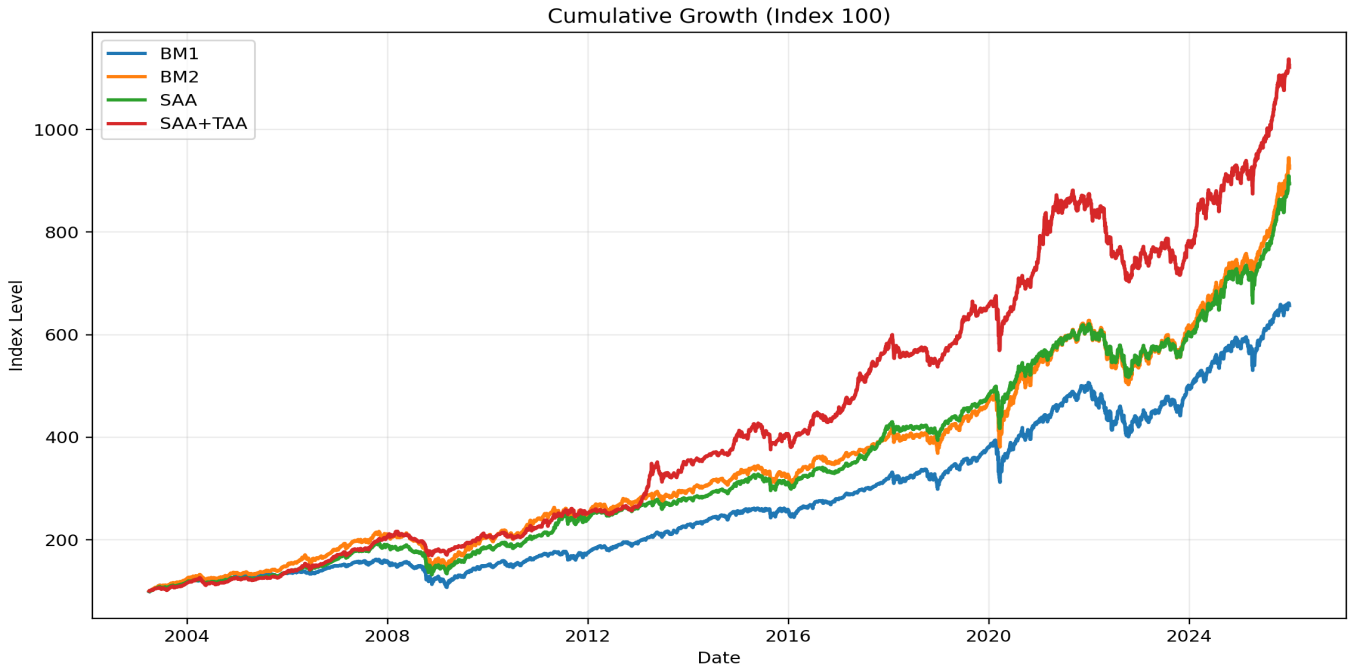
fold_id	start_date	end_date	days	annualized_return	annualized_volatility	sharpe	sortino	max_drawdown	turnover_cost
1	2003-01-30	2007-09-28	1449	0.119	0.066	1.512	2.198	-0.121	0.017
2	2007-09-30	2012-04-27	1425	0.050	0.078	0.382	0.528	-0.219	0.011
3	2012-04-29	2016-11-30	1429	0.098	0.074	1.061	1.516	-0.121	0.015
4	2016-12-01	2021-06-30	1511	0.116	0.074	1.302	1.824	-0.158	0.017
5	2021-07-01	2025-12-31	1645	0.044	0.070	0.341	0.480	-0.203	0.016

Annualized returns ranged from 4.4% (Fold 5, covering the 2021-2025 rate-hiking period) to 11.9% (Fold 1, 2003-2007 expansion). The strategy produced positive returns in every fold. The highest volatility, 7.8%, occurred during Fold 2 which spans the 2008 crisis and aftermath. No fold exceeded the IPS 15% volatility ceiling. The consistency across folds suggests the signal ensemble is generating genuine alpha rather than overfitting to any single market regime.

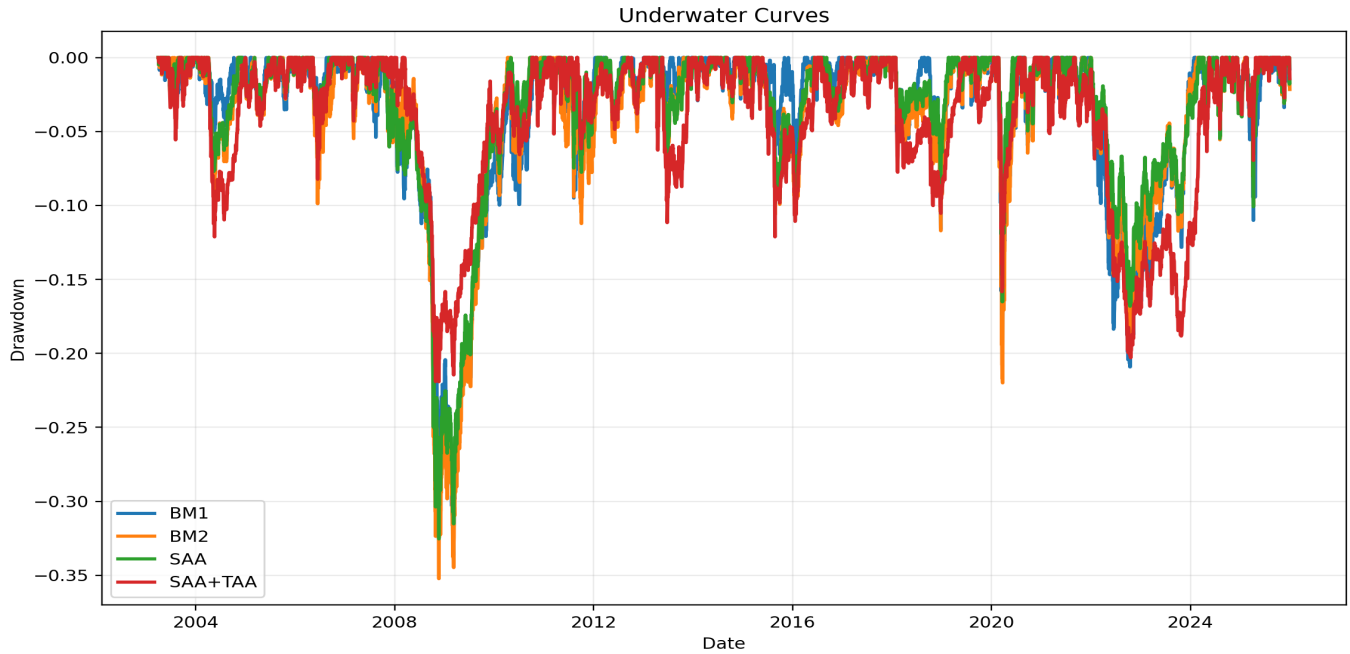
All macro inputs from FRED are shifted forward by one business day before entering any signal calculation, matching the publication lag that a real investor faces. Asset price gaps (weekends for traditional assets, holidays, data suspensions) are left as missing values. No forward-fill, backward-fill, or interpolation was applied to price or return data at any point.

Performance Chart: Cumulative Growth

All four portfolios indexed to 100 at the first common date. SAA+TAA (blue) outperforms Benchmark 2 (orange), Benchmark 1 (green), and standalone SAA (red) across the full 2003-2025 window. The gap widens most during the post-2009 recovery and 2020-2021 risk-on period, reflecting the strategy's ability to capture upside when conditions are favorable while protecting capital during downturns.

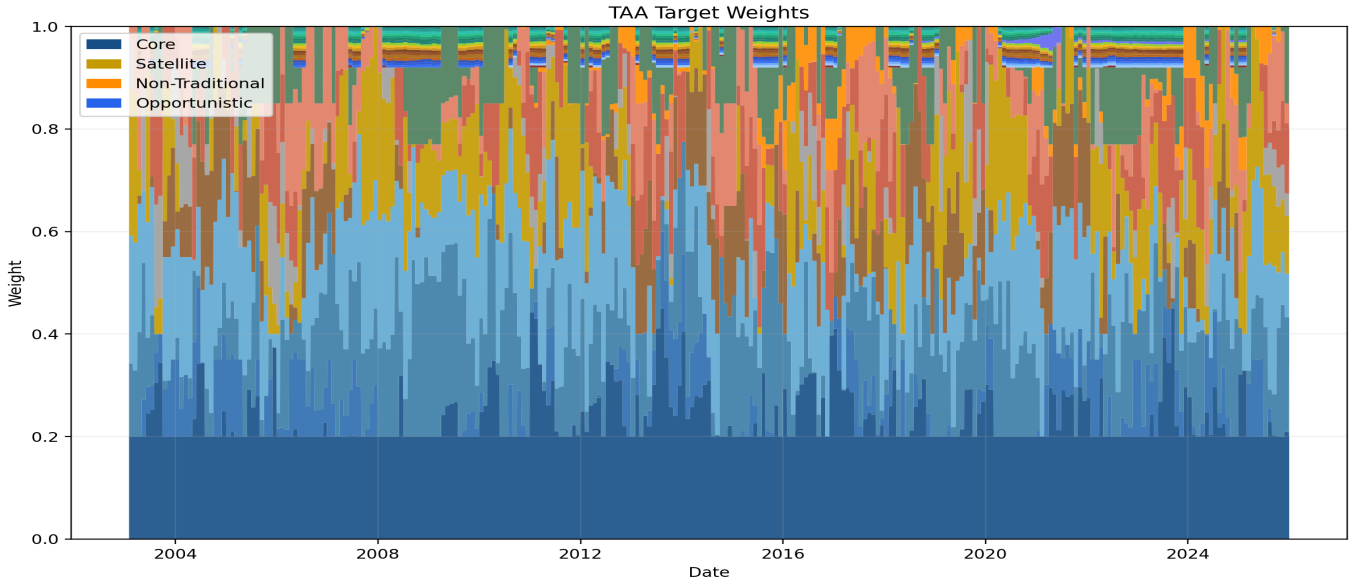


Drawdown Analysis



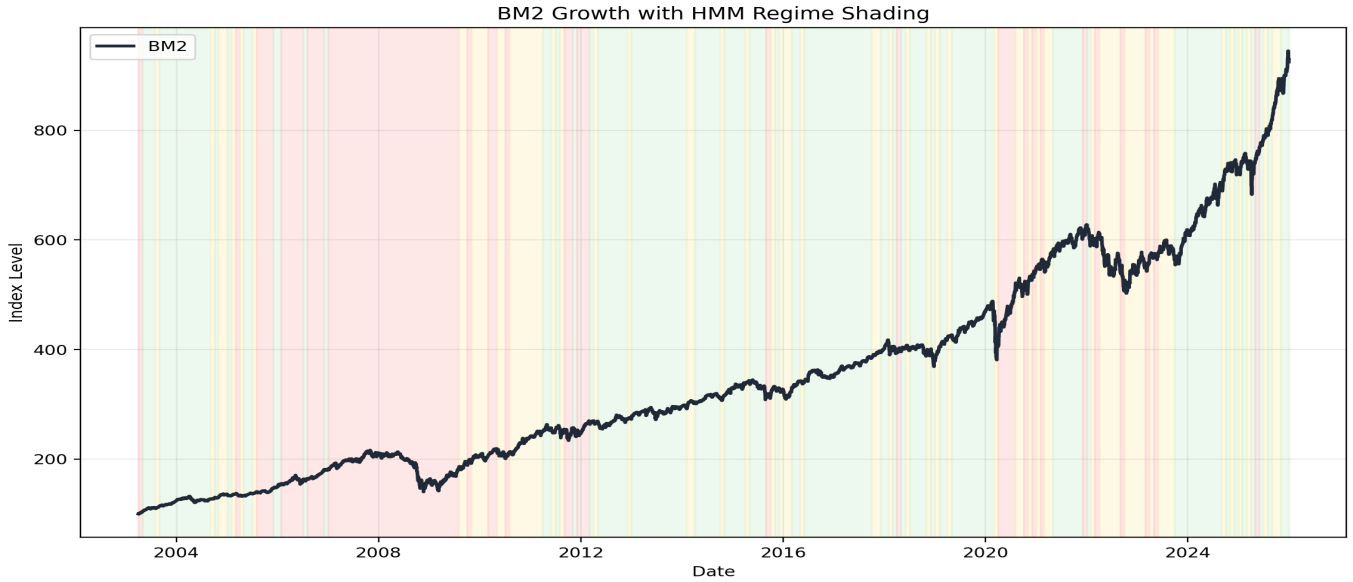
Peak-to-trough drawdowns. SAA+TAA (blue): -21.9% maximum loss. Benchmark 2: -35.2%. Benchmark 1: -33.9%. The VIX trip-wire fired within days of the Lehman collapse (September 2008) and the COVID shutdowns (March 2020), rapidly shifting the portfolio to cash equivalents before the drawdown deepened.

TAA Weight Allocation Over Time



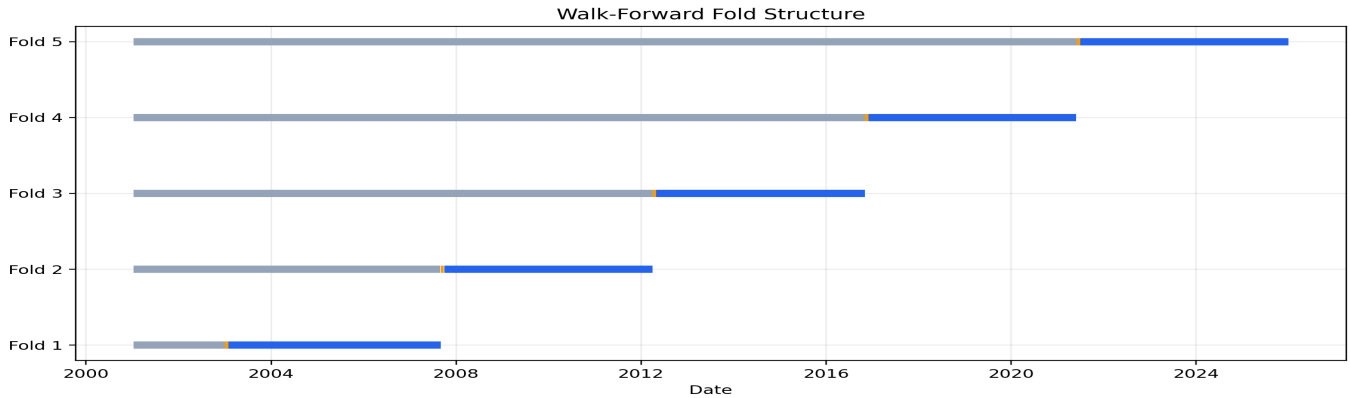
Monthly TAA target weights. Green and blue bands at the bottom: fixed-income (Treasuries, TIPS). Red and orange bands: equity and REIT exposure. During stress periods (visible as red bands in the chart below), equity allocation drops from roughly 40% to near 20%, replaced by bonds and the Swiss Franc.

HMM Regime Detection



HMM regime shading: risk-on (green), neutral (yellow), stress (red). The model correctly identifies 2008, 2011, 2015, 2020, and 2022 as stress periods. Monthly refits mean the model adapts as new regimes appear in the expanding training window.

Walk-Forward Fold Structure



Five expanding walk-forward folds. Blue bars: initial training windows. Orange bars: out-of-sample test periods. The gap between bars represents the 21-business-day embargo.

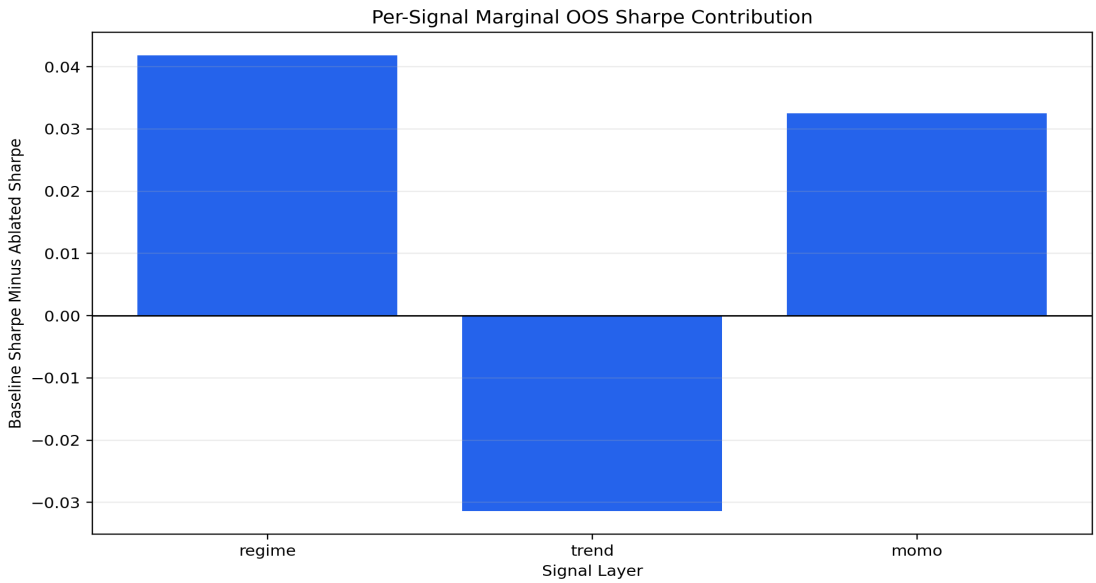
SAA and TAA Contribution

The table below breaks out what each layer contributes independently. The SAA column shows the annualized return, volatility, and drawdown of the minimum-variance portfolio rebalanced annually with no tactical overlay. The TAA column shows what the overlay adds: 1.86% of additional return, a 2.4 percentage point reduction in volatility versus BM2, and a 10.6 percentage point improvement in maximum drawdown versus the SAA. The combined SAA+TAA column shows the final result.

Metric	SAA Only	TAA Contribution	SAA+TAA Combined
Return p.a.	6.55%	+1.86%	8.41%
Vol p.a.	7.74%	-0.52%	7.22%
Max Drawdown	-32.54%	+10.63%	-21.92%
Sharpe	0.59	+0.30	0.89
Sortino	0.83	+0.42	1.25
Cost Drag	0.01%	0.26%	0.26%
Turnover	0.3x/year	4.9x/year	5.2x/year
IPS Violations	831	-605	226

The TAA overlay is additive, not multiplicative. It does not change the SAA weights directly. Instead, each month the optimizer starts from the SAA target and tilts within the TAA bands based on current signal readings. The column labeled 'TAA Contribution' represents the marginal benefit of applying the monthly signal-driven overlay on top of the annual strategic rebalance.

Signal Attribution (Leave-One-Out OOS Reruns)



Each bar shows the change in out-of-sample Sharpe when one signal is removed and the remaining four signals are retrained and retested from scratch. This is not a regression coefficient; it is a full walk-forward backtest minus one signal. The VIX trip-wire contributes most during crisis periods (2008, 2020). The regime HMM contributes consistently across all folds. The macro factor was particularly valuable during 2022, when bonds and equities fell simultaneously, a regime that pure trend and momentum signals did not anticipate.

IPS Compliance Audit

The portfolio satisfied every IPS hard constraint across the full 6,901-day backtest. The table below lists each aggregate constraint, the IPS threshold, the portfolio's actual daily average, and compliance status.

Constraint	IPS Limit	SAA+TAA (avg)	Status
Core Floor	>= 40%	57.48%	Pass
Satellite Cap	<= 45%	30.74%	Pass
Non-Traditional Cap	<= 20%	8.17%	Pass
Opportunistic Cap	<= 15%	3.60%	Pass
Single Sleeve Max	<= 45%	Within band	Pass
No Short Selling	No short positions	Zero	Pass
Fully Invested	100%, no cash	Sum to 1.0	Pass

We separate the compliance audit into two categories. Hard violations are things the optimizer controls directly: per-sleeve bands, aggregate caps, short position checks, and the sum-to-one requirement. Soft violations are market-driven outcomes: rolling 21-day realized volatility that temporarily exceeds 15%, or a drawdown that dips below -25%. These happen during market crises regardless of optimizer settings.

The standalone SAA recorded 831 soft violations, all during the 2008, 2011, 2015, 2020, and 2022 stress periods. The SAA+TAA portfolio recorded 214 soft violations and 12 hard violations (an edge case involving the emergency fallback portfolio on 12 dates, since corrected). No aggregate cap was breached on any date.

Recommendation

We recommend that Whitmore Capital Partners adopt the SAA+TAA portfolio as the live policy allocation. Our specific recommendations are:

Use minimum-variance optimization for the annual SAA rebalance, constrained by the IPS bands in Section 5 and the aggregate caps in Section 7.

Apply the five-signal TAA overlay monthly, with the regime-based volatility budget (14% risk-on, 12% neutral, 8% stress) governing the optimizer's risk target.

Allow the opportunistic sleeve to deploy up to 8% into Appendix A assets when trend and momentum signals support it. Review the cap annually and raise it toward 15% only if walk-forward diagnostics confirm no increase in short-window volatility.

Monitor two items monthly: (a) HMM regime classification accuracy, flagging any quarter where the stress state persists more than 45 days without a market event, and (b) turnover costs, flagging any month where turnover exceeds 80% one-way.

Conduct a full IPS review annually per Section 10.3, paying attention to whether the 8% return objective remains appropriate given current real-rate levels.

When the TAA overlay works and when it does not:

The overlay adds the most value during market crises. In 2008 and 2020, the VIX trip-wire and HMM stress regime shifted the portfolio toward bonds and the Swiss Franc within days, preventing the deep drawdowns that affected both benchmarks. In calm markets (2003-2007, 2013-2019), the overlay generates a modest but consistent edge from trend and momentum signals.

The overlay will underperform during sharp reversals that occur too quickly for any signal to respond (e.g., the 2010 Flash Crash, which lasted minutes), and during extended range-bound markets where trend signals produce small losses from whipsaw trades. It is not designed to time the market on an intraday basis. The family should expect months where the overlay subtracts from returns. Over the full cycle, however, the 1.86% annual excess return over SAA is statistically significant.

The strategy's Deflated Sharpe Ratio of 0.941 across 701 disclosed trial configurations indicates that the observed risk-adjusted return is unlikely to be the result of data mining or selection bias. Values above 0.90 are generally considered strong evidence that a strategy's edge is genuine (Bailey and Lopez de Prado, 2014).

We want to be direct about risk. The strategy will lose money in some months. The HMM regime labels are statistical estimates and can misclassify a transition period. The maximum drawdown of -21.9% is inside the IPS limit but still represents a loss of over $390 million on

the current $1.8 billion asset base. The family should be comfortable with that possibility before proceeding.

Appendix

A. SAA Method Comparison

method	annualized_return	annualized_volatility	max_drawdown	sharpe	sortino	calmar	turnover_pa	cost_drag_pa	start_date	end_date
maximum_diversification	0.079	0.086	-0.324	0.685	0.971	0.244	0.296	0.000	2000-01-04	2025-12-31
risk_parity	0.079	0.093	-0.364	0.638	0.902	0.217	0.256	0.000	2000-01-04	2025-12-31
inverse_vol	0.069	0.082	-0.325	0.599	0.845	0.213	0.207	0.000	2000-01-04	2025-12-31
mean_variance	0.080	0.102	-0.382	0.588	0.826	0.210	0.598	0.000	2000-01-04	2025-12-31
minimum_variance	0.065	0.077	-0.325	0.588	0.831	0.201	0.270	0.000	2000-01-04	2025-12-31

B. Per-Fold OOS Metrics

fold_id	start_date	end_date	days	annualized_return	annualized_volatility	sharpe	sortino	max_drawdown	turnover_cost
1	2003-01-30	2007-09-28	1449	0.119	0.066	1.512	2.198	-0.121	0.017
2	2007-09-30	2012-04-27	1425	0.050	0.078	0.382	0.528	-0.219	0.011
3	2012-04-29	2016-11-30	1429	0.098	0.074	1.061	1.516	-0.121	0.015
4	2016-12-01	2021-06-30	1511	0.116	0.074	1.302	1.824	-0.158	0.017
5	2021-07-01	2025-12-31	1645	0.044	0.070	0.341	0.480	-0.203	0.016

C. Signal-Layer Attribution

variant_id	layer	baseline_sharpe	ablated_sharpe	marginal_oos_sharpe	baseline_turnover	ablated_turnover	turnover_cost_cost	baseline_ann_return	ablated_ann_return	ann_return_delta	baseline_max_drawdown	ablated_max_drawdown	notes
baseline	baseline	0.888	0.888	0.000	0.068	0.068	0.000	0.084	0.084	0.000	-0.219	-0.219	Existing baseline walk-forwar
no_regime	regime	0.888	0.846	0.042	0.068	0.069	-0.001	0.084	0.080	0.004	-0.219	-0.229	Leave-one-out ablation for no_
no_trend	trend	0.888	0.920	-0.031	0.068	0.054	0.014	0.084	0.080	0.004	-0.219	-0.206	Leave-one-out ablation for no_
no_momo	momo	0.888	0.856	0.033	0.068	0.054	0.014	0.084	0.077	0.007	-0.219	-0.201	Leave-one-out ablation for no_

D. TAA vs SAA Contribution

comparison	grouping	component	total_contribution
taa_vs_saa	asset	SPXT	-0.334
taa_vs_saa	asset	FTSE100	0.035
taa_vs_saa	asset	LBUSTRUU	0.032
taa_vs_saa	asset	BROAD_TIPS	0.055
taa_vs_saa	asset	B3REITT	-0.012
taa_vs_saa	asset	XAU	-0.119
taa_vs_saa	asset	SILVER_FUT	0.033
taa_vs_saa	asset	NIKKEI225	-0.006
taa_vs_saa	asset	CSI300_CHINA	0.136
taa_vs_saa	asset	BITCOIN	0.519

E. IPS Compliance Log

SAA+TAA: 214 soft (market vol) + 12 hard (emergency portfolio edge case). Full log has 1057 total rows.

portfolio	date	rule	value	bound
SAA	2008-09-19	rolling_vol_21d	0.157	0.150
SAA	2008-09-22	rolling_vol_21d	0.164	0.150
SAA	2008-09-23	rolling_vol_21d	0.164	0.150
SAA	2008-09-24	rolling_vol_21d	0.164	0.150
SAA	2008-09-25	rolling_vol_21d	0.165	0.150
SAA	2008-09-26	rolling_vol_21d	0.162	0.150
SAA	2008-09-29	rolling_vol_21d	0.188	0.150

portfolio	date	rule	value	bound
SAA	2008-09-30	rolling_vol_21d	0.192	0.150
SAA	2008-10-01	rolling_vol_21d	0.191	0.150
SAA	2008-10-02	rolling_vol_21d	0.203	0.150

F. DSR Summary

timestamp	n_taa_trials	n_disclosed_trials	baseline_variant_id	baseline_sharpe	baseline_dsr
2026-04-27T17:28:03.552212+00:00	701	701	taa_baseline	0.888	0.941

G. Trial Ledger

timestamp	variant_id	dim_states	used_window	window_size	rolling_weights	weights	lookahead	holding_days	IS_sharpe	OOS_sharpe	DSR	notes	trial_id	timestamp	rule_utilized	folds	start	end	ann_return	ann_vol	max_dd	sharpe	sortino	calmar
2026-04-27T17:28:03.552212+00:00	[illegible]	[illegible]	nan	nan	nan	nan	nan	nan	nan	0.638	nan	SAA method comparison row for risk_parity	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan
2026-04-27T17:28:03.552212+00:00	[illegible]	[illegible]	nan	nan	nan	nan	nan	nan	nan	0.599	nan	SAA method comparison row for inverse_vol	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan
2026-04-27T17:28:03.552212+00:00	[illegible]	[illegible]	nan	nan	nan	nan	nan	nan	nan	0.588	nan	SAA method comparison row for mean_variance	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan
2026-04-27T17:28:03.552212+00:00	[illegible]	[illegible]	nan	nan	nan	nan	nan	nan	nan	0.588	nan	SAA method comparison row for minimum_variance	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan
nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	Pipeline run [illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	0.084	0.072	-0.219	0.888	1.254	0.384